

Task Management API - Stage 5:

OpenAPI 3.0 Blueprint & Swagger UI

Interactive Verification

Overview

This repository hosts a RESTful Task Management API built with Node.js and Express in VS Code on Windows as part of the FlyRank Internship Backend Engineering Track.

Stage 5 introduces interactive API documentation and testing capabilities by integrating an **OpenAPI 3.0 blueprint** (openapi.json) and mounting **Swagger UI** (swagger-ui-express) at the /docs endpoint. This allows developers and system administrators to discover, inspect, and execute interactive CRUD operations directly through a web interface.

Technical Stack & Environment

- **Runtime:** Node.js (v24.x)
- **Framework:** Express.js
- **API Specification Standard:** OpenAPI 3.0 Specification (openapi.json)
- **Documentation UI Engine:** swagger-ui-express
- **Middleware:** express.json() (Deep Packet Inspection / Body Parser)
- **Data Store:** In-Memory Volatile Array (RAM Cache)
- **Environment:** VS Code Integrated Terminal (Windows PowerShell pwsh)
- **Interactive Endpoint:** <http://localhost:3000/docs/>
- **Version Control:** Git

Architectural Breakdown: Stage 5 Concepts

1. Specification Decoupling (openapi.json)

The API blueprint is defined in a separate openapi.json contract file. This separates the runtime route handlers (server.js) from the interface documentation, ensuring that third-party integrations, frontend teams, and automated API clients have a formal single source of truth.

2. Path & Parameter Synchronization

To ensure exact alignment between Express routing and Swagger UI, paths are explicitly mapped:

- **Collection Scope (/tasks):**
 - GET /tasks — Retrieve all tasks in active inventory.
 - POST /tasks — Register a new task with JSON payload validation.

- **Item Scope (/tasks/{id}):**
 - GET /tasks/{id} — Lookup specific task by dynamic path ID.
 - PUT /tasks/{id} — Update task attributes (title, done).
 - DELETE /tasks/{id} — Remove task from active memory space.

3. Sysadmin / Enterprise Analogy

Mounting Swagger UI on an Express application is equivalent to deploying an **API Gateway Developer Portal** (such as AWS API Gateway Portal or Azure API Management). It transforms backend socket routes into an interactive control panel where sysadmins and developers can perform operational diagnostics without manually drafting raw curl commands.

Interactive Verification Lifecycle (Swagger UI)

All CRUD operations were verified through the Swagger UI web portal at <http://localhost:3000/docs/>.

1. Launch Server Daemon

```
node server.js
```

Daemon Output:

Task API Daemon running on <http://localhost:3000>
 Swagger UI available at <http://localhost:3000/docs>

2. Step-by-Step Interactive Test Log

Step 1: Create Task (POST /tasks)

- **Swagger Action:** Expanded POST /tasks → Clicked **Try it out**.
- **Payload Submitted:**

```
{
  "title": "Automate Kenyan SME invoices"
}
```
- **Status Code:** 201 Created
- **Response Body:**

```
{
  "id": 4,
  "title": "Automate Kenyan SME invoices",
  "done": false
}
```

Step 2: Query Task (GET /tasks/{id})

- **Swagger Action:** Expanded GET /tasks/{id} → Set path parameter id = 4.
- **Status Code:** 200 OK
- **Response Body:**

```
{
  "id": 4,
  "title": "Automate Kenyan SME invoices",
  "done": false
}
```

Step 3: Mutate State (PUT /tasks/{id})

- **Swagger Action:** Expanded PUT /tasks/{id} → Set id = 4.
- **Payload Submitted:**

```
{
  "title": "Automate Kenyan SME invoices",
  "done": true
}
```
- **Status Code:** 200 OK
- **Response Body:**

```
{
  "id": 4,
  "title": "Automate Kenyan SME invoices",
  "done": true
}
```

Step 4: Delete Task (DELETE /tasks/{id})

- **Swagger Action:** Expanded DELETE /tasks/{id} → Set id = 4.
- **Status Code:** 204 No Content
- **Verification:** Re-querying GET /tasks/4 returns 404 Not Found, confirming complete record purge from system memory.

Service Map & Endpoint Reference

HTTP Method	Route Path	Path Param	Request Body	Success Code	Error Codes	Operation Summar
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GET	/tasks	None	None	200 OK	—	Retrieve full task list
POST	/tasks	None	{"title": string}	201 Created	400 Bad Request	Register a new task
GET	/tasks/{id}	id (int)	None	200 OK	404 Not Found	Query task by Primary Key ID
PUT	/tasks/{id}	id (int)	{"title": string, "done": bool}	200 OK	400 Bad Request, 404 Not Found	Update task state or properties
DELETE	/tasks/{id}	id (int)	None	204 No Content	404 Not Found	Purge task from active RAM

Version Control Checkpoint

Save and commit all Stage 5 OpenAPI & Swagger documentation changes:

```
git add .
```

```
git commit -m "Stage 5: OpenAPI 3.0 specification and interactive Swagger UI verification"
```